



Bulk phase biochemistry of PIF1 and RecQ4 family helicases

**Prasangi Rajapaksha, Robert H. Simmons III, Spencer J. Gray,
David J. Sun, Phoebe Nguyen, David G. Nickens,
and Matthew L. Bochman***

Molecular & Cellular Biochemistry Department, Indiana University, Bloomington, IN, United States

*Corresponding author: e-mail address: bochman@iu.edu

Contents

1. Introduction	170
2. Before you begin	172
3. Materials and equipment	173
3.1 General equipment	173
4. Step-by-step method details	175
4.1 Recombinant protein over-expression and purification	175
4.2 DNA substrate preparation	178
4.3 DNA binding assay	181
4.4 ATPase assay	182
4.5 Helicase assay	184
4.6 DNA annealing assay	186
4.7 Strand exchange assay	187
References	187

Abstract

DNA helicases are involved in nearly all facets of genome integrity, and in humans, mutations in helicase-encoding genes are often linked to diseases of genomic instability. Two highly studied and evolutionarily conserved helicase families are the PIF1 and RecQ helicases. Enzymes in these families have known roles in DNA replication, recombination, and repair, as well as telomere maintenance, DNA recombination, and transcription. Although genetics, structural biology, and a variety of other techniques have been used to study these helicases, ensemble analyses of their basic biochemical activities such as DNA binding, ATP hydrolysis, and DNA unwinding have made significant contributions to our understanding of their physiological roles. Here, we present general methods to generate recombinant proteins from both helicase families, as well as standard biochemical assays to investigate their activities on DNA.



1. Introduction

DNA helicases can be grouped into families and superfamilies based on their conserved sequence motifs, oligomeric state, and polarity of action on DNA (Berger, 2008). For instance, PIF1 family helicases all contain a 21-amino acid signature sequence (Bochman, Judge, & Zakian, 2011; Bochman, Sabouri, & Zakian, 2010) between their conserved helicase motifs II and III (Hall & Matson, 1999), are monomeric in solution, and unwind DNA in a 5'-3' direction, making them members of Superfamily IB (SF-IB). In contrast, as members of SF-IIA, RecQ family helicases all contain a RecQ C-terminal domain (RQC) adjacent to their central helicase domain, are monomeric in solution, and display 3'-5' polarity (Berger, 2008). Although these basic features hold true for all of the studied family members, species-specific differences often occur in the form of additional domains. For example, one of the human RecQ helicases, WRN, includes a 3'-5' exonuclease domain that isn't found in yeast or bacterial RecQs (Bernstein, Gangloff, & Rothstein, 2010), and eukaryotic PIF1 helicases often contain N- and C-terminal domains flanking their conserved helicase domain that are not found in bacterial PIF1s (Andis, Sausen, Alladin, & Bochman, 2018; Bochman et al., 2011, 2010). Based on this protein-to-protein variability, members of both the PIF1 and RecQ helicase families have evolved to perform a variety of distinct functions *in vivo*.

The first PIF1 helicase, known simply as Pif1 or ScPif1, was identified 40 years ago in *Saccharomyces cerevisiae*, where it is involved in mitochondrial DNA replication and recombination (Bochman et al., 2010; Foury & Dyck, 1985). Since then, an additional family member called Rrm3 was discovered in yeast, and various functions for the *S. cerevisiae* PIF1 helicases have been described, such as telomere maintenance, nuclear DNA replication, DNA break repair, and G-quadruplex structure resolution (reviewed in Muellner & Schmidt, 2020). Additionally, PIF1 helicases have been identified and studied in other organisms, such as Pfh1 from the fission yeast *Schizosaccharomyces pombe*, multiple family members in the kinetoplastid parasite *Trypanosoma brucei*, the cancer-linked human PIF1, and the modular PIF1 helicases from thermostable bacteria (Andis et al., 2018; Bochman et al., 2010; Muellner & Schmidt, 2020). Such studies have shed light on the physical and genetic interactomes of these helicases (Stundon & Zakian, 2015; Zybailov et al., 2015), their regulation by post-translational

modifications (Ononye, Sausen, Balakrishnan, & Bochman, 2020; Ononye, Sausen, Bochman, & Balakrishnan, 2021; Rossi, Ajazi, Carotenuto, Foiani, & Giannattasio, 2015; Vasanovich, Harrington, & Makovets, 2014), and the contributions of their unique non-helicase domains (Andis et al., 2018; Bessler & Zakian, 2004; Nickens, Sausen, & Bochman, 2019).

The RecQ family helicases share a similar lengthy and impressive history as the focus of many researchers (Chu & Hickson, 2009). Originally identified in *Escherichia coli*, the RecQs are now known to be highly conserved throughout evolution, with more complex organisms generally encoding an increasing number of family members. For instance, simple eukaryotes like *S. cerevisiae* contain two RecQ helicases (Hrq1 and Sgs1), while the human genome encodes five (RECQL1, BLM, WRN, RECQL4, and RECQL5) (Bochman, 2014). In this case, Hrq1 and Sgs1 are functionally homologous to RECQL4 and BLM, respectively. Similar to the PIF1 helicases, RecQ helicases function in a variety of processes *in vivo*, from DNA replication and transcription to DNA recombination and repair (Bachrati & Hickson, 2008; Bernstein et al., 2010; Bohr, 2008; Chu & Hickson, 2009; Gupta & Schmidt, 2020).

We have recently focused on the RecQ4 subfamily of RecQ helicases because they are the only RecQs found in all three domains of life and play important but often undefined roles in genome integrity. Additionally, mutations in human RECQL4 cause three distinct diseases (reviewed in Rogers, Simmons, Fluhler Thornburg, Buehler, & Bochman, 2020). RecQ4 subfamily helicases also genetically, physically, and functionally interact with other RecQs and PIF1 family helicases (Bochman, Paeschke, Chan, & Zakian, 2014; Nickens, Rogers, & Bochman, 2018; Rogers, Sanders, et al., 2020; Sanders, Nguyen, Rogers, & Bochman, 2020; Simmons, Rogers, & Bochman, 2021). In *S. cerevisiae*, these interactions impact telomerase regulation and the cellular responses to various forms of DNA damage.

Here, we provide detailed methods for the construction of DNA substrates, the purification of recombinant proteins, and biochemical assays to measure the activities of PIF1 family and RecQ4 subfamily helicases *in vitro*. These bulk phase assays can be used to test hypotheses about the functions of the helicases based on *in vivo* experiments. Conversely, they can act as a starting point to learn about the basic activities of the helicases and generate testable hypotheses for cell-based investigations. Indeed, we have recently used this type of give and take to begin to unravel the mechanism of DNA inter-strand crosslink (ICL) repair in *S. cerevisiae*, which involves the Hrq1 helicase (Rogers, Lee, et al., 2020).



2. Before you begin

Our best advice to those interested in performing the types of biochemical assays described below, especially for those who are new to protein purification and *in vitro* assays in general, is to review the Kornberg Commandments ([Kornberg, 2003](#)). Commandments IV (Do not waste clean thinking on dirty enzymes) and V (Do not waste clean enzymes on dirty substrates) are particularly germane here. A beautiful DNA substrate preparation will be ruined by a helicase stock that is contaminated with an expression host nuclease, and likewise, a crystallography-worthy helicase preparation can only tell you so much if your DNA substrate is not pristine. In other words, biochemistry is often slow and front-loaded by the necessity to troubleshoot protein and substrate purification problems, but rushing ahead with so-so reagents often just wastes more time.

In our experience, protein over-expression and purification is usually the limiting step in completing these investigations. Many of the PIF1 and RecQ helicases, particularly those found in eukaryotes, are large proteins and/or contain sizeable domains that are predicted to be natively disordered. These qualities make over-expression difficult in *E. coli*, for instance, but baculovirus-infected insect cell culture tends to perform much better. Although the latter requires more time and more expensive growth media, the time and money are worth it when a single 1-L insect cell preparation yields orders of magnitude more protein than tens of liters of *E. coli* cells over-expressing your protein ([Nickens & Bochman, 2021](#)).

Finally, it should be noted that there are limitations to the ensemble assays described below. Most are endpoint experiments and do not yield kinetics data. Even those that can be performed in a time course fashion are still limited by how quickly time points can be taken. Further, bulk phase assays can lack sufficient resolution to yield detailed mechanistic data, a situation that we encountered when investigating ICL repair by Hrq1 ([Rogers, Lee, et al., 2020](#)). In such cases, it is advisable to consider single-molecule approaches that enable real-time data collection and high-resolution observation of the reaction being investigated. Regardless, the methods described herein are an excellent starting point that can guide the establishment of proper single-molecule experiments. They are also quite versatile—being amenable to order-of-addition investigations, reagent titrations vs time course comparisons, and supplementation with various reagents like small molecule inhibitors—with only minimal and straight-forward modifications of the protocols necessary.



3. Materials and equipment

3.1 General equipment

- Vertical polyacrylamide gel electrophoresis equipment (CBS Scientific Vertical Mini-gel System)
- Standing incubators: 27, 30, and 37 °C
- Shaking incubators capable of heating ($\geq 37^\circ\text{C}$) and cooling ($\leq 18^\circ\text{C}$)
- Centrifuges (micro, table-top, and super-speed) capable of cooling to 4 °C
- Sonicator or alternative method for bacterial cell lysis
- Medium-pressure liquid chromatography instrument (AKTA Pure)
- Liquid nitrogen Dewar
- LI-COR Odyssey DLx
- pH meter

3.1.1 Strains

- *E. coli* strain Rosetta 2(DE3)pLysS (F^- *ompT* *hsdS_B*(r_B^- m_B^-) *gal dcm* (DE3) pLysSRARE2 (Cam^R))
- *Spodoptera frugiperda* cell line Sf9

3.1.2 Reagents and chemicals

- Protease inhibitor mix: 600 nM leupeptin, 2 μM pepstatin A, 2 mM benzamidine, and 1 mM phenylmethanesulfonyl fluoride (PMSF). Commercially available protease inhibitor mixes are also appropriate
- NaCl
- Glycerol
- Lysozyme
- β -Mercaptoethanol
- HEPES
- EDTA
- Tris
- HCl
- NaOH
- NaOAc
- MgCl_2
- Triton-X100
- NP-40 or NP-40 substitute
- $\text{Mg}(\text{OAc})_2$
- $\text{Zn}(\text{OAc})_2$

- NH_4OAc
- Dithiothreitol (DTT)
- Tween-20
- SDS
- Proteinase K
- 100mM ATP (pH7.0): dissolve 1.2g ATP in 15mL Milli-Q H_2O , adjust the pH to 7.0 with 0.1 N NaOH, bring the volume to 20 mL with H_2O , make 1-mL aliquots, and store them at -80°C
- 10mM phosphoenolpyruvic acid (PEP): dissolve 20.6 mg PEP in 10 mL Milli-Q H_2O , make 500- μL aliquots, and store them at -80°C
- 10mM NADH: dissolve 43mg NADH in 10mL Milli-Q H_2O , make 400- μL aliquots, and store them at -80°C
- SDS-Proteinase K: Dissolve Proteinase K to 4mg/mL in 50mM Tris-HCl (pH8.0) and 3mM CaCl_2 . Make small aliquots, and store them at -80°C

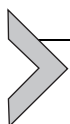
3.1.3 Homemade buffers

- Pif1 lysis buffer: 50mM sodium phosphate buffer (pH7.5), 300mM NaCl, 5mM β -mercaptoethanol, 10% (w/v) glycerol, protease inhibitor mixture, and 0.5mg/mL lysozyme
- Pif1 IEX buffer: 50mM sodium phosphate buffer (pH6.8), 10% (w/v) glycerol, 0.5mM EDTA (pH8.0), and 4mM β -mercaptoethanol
- Pif1 storage buffer: 25mM HEPES (pH7.5), 150mM NaCl, 2mM β -mercaptoethanol, 0.1mM EDTA (pH8.0), and 30% (w/v) glycerol
- L buffer: 40mM Tris-HCl (pH8.0), 500mM NaOAc, 1.1mM MgCl_2 , 0.1mM EDTA, 0.1% (w/v) Triton X-100, 0.2% (w/v) NP-40, 10% (w/v) glycerol, and 0.1mM PMSF
- $2\times$ resuspension buffer: 100mM Na-HEPES (pH8.0), 10% (w/v) glycerol, 300mM NaOAc (pH8.0), 10mM $\text{Mg}(\text{OAc})_2$, and 0.1% Tween-20
- Gel filtration buffer: 25mM Na-HEPES (pH7.5), 5% glycerol, 300mM NaCl, 5mM MgCl_2 , and 0.05% Tween-20
- Hrq1 storage buffer: 25mM Na-HEPES (pH8.0), 30% (w/v) glycerol, 300mM NaOAc (pH7.6), 25mM NaCl, 5mM $\text{Mg}(\text{OAc})_2$, 1mM DTT and 0.1% (v/v) Tween-20
- $2\times$ annealing buffer: 40mM Tris-HCl (pH8.0), 8% glycerol, 0.2mM EDTA, 80 $\mu\text{g/mL}$ BSA, 20mM DTT, and 20mM $\text{Mg}(\text{OAc})_2$
- $1\times$ ATPase buffer: 25mM Na-HEPES (pH8.0), 5% (w/v) glycerol, 50mM NaOAc (pH7.5), 150mM NaCl, and 0.01% (v/v) NP-40

- 2 × binding buffer: 50 mM Na-HEPES (pH 8.0), 10% (w/v) glycerol, 100 mM NaOAc (pH 7.6), 300 μM NaCl, 15 mM Mg(OAc)₂, and 0.02% Tween-20
- 6 × nonfluorescent loading buffer: 50 mM Tris (pH 8.0), 25% (w/v) glycerol, and 0.025% Orange G
- 1 × TG running buffer: 25 mM Tris and 185 mM glycine (pH 8.8)
- 1 × TBE running buffer: 90 mM Tris base, 90 mM boric acid, and 2 mM EDTA (pH 8.0)

3.1.4 Oligonucleotides

We exclusively purchase our modified and unmodified oligonucleotides from Integrated DNA Technologies (IDT). The price per nucleotide at the smallest production scale (≤ 60 nt) is competitive and sufficient for hundreds of reactions for most of the substrates that we typically use *in vitro*. In most cases, standard desalting is the only purification required, though when using substrates in gels with single-nucleotide resolution (e.g., Nickens et al., 2018, 2019; Rogers, Lee, et al., 2020), PAGE purification is necessary. Further, IDT offers a wide variety of 5', 3', and internal modifications, including the near-infrared (NIR) fluorophores IRDye[®] 700 and 800 suggested below.



4. Step-by-step method details

4.1 Recombinant protein over-expression and purification

4.1.1 PIF1 family helicases

PIF1 family helicases have been investigated from a variety of organisms (Bochman et al., 2010; Muellner & Schmidt, 2020), and we recently published a detailed protocol (Nickens & Bochman, 2021) for the over-expression and purification of *S. cerevisiae* Pif1 and mutants thereof (Nickens et al., 2018, 2019; Ononye, Sausen, et al., 2020). Here, we include a protocol for the generation of recombinant PIF1 helicases from bacteria (Andis et al., 2018), which is also amenable to *S. cerevisiae* Pif1.

1. Transform a PIF1 expression plasmid such as pMB427 (Andis et al., 2018) into Rosetta 2(DE3) pLysS (Novagen) *E. coli* cells. Select for transformants on LB medium supplemented with 100 μg/mL ampicillin and 34 μg/mL chloramphenicol at 37 °C.
2. Inoculate one or more 5-mL cultures of LB medium supplemented with the same antibiotics with a match head-sized swab of cells from

the transformation plate, and grow for ~6 h at 30 °C with aeration. Only fresh transformants should be used as we have observed little-to-no expression from transformants stored on plates for extended periods of time or from glycerol stocks of expression strains carrying the pSUMO-Pif1 plasmid.

3. Back-dilute these starter cultures 1:100 into ZYP-5052 autoinduction medium (Studier, 2005) containing $1 \times$ trace metals mix, 100 $\mu\text{g/mL}$ ampicillin, and 34 $\mu\text{g/mL}$ chloramphenicol, and incubate at 22 °C with agitation to $\text{OD}_{600} > 3$ (15–18 h). Unless otherwise stated, all subsequent steps should be performed at 4 °C or on ice using instruments and equipment that is likewise chilled.
4. Harvest the cells by centrifugation at $5500 \times g$ for 10 min, decant the growth medium, weigh the pellet(s), and freeze the cells at -80°C prior to lysis.
5. Resuspend the cells by gently stirring in 4 mL/g cell pellet $1 \times$ resuspension buffer supplemented with protease inhibitor mix and 20 $\mu\text{g/mL}$ DNase I.
6. Lyse cells using a high-pressure cell cracker operated at >1000 psi, clarify the lysate by centrifugation at $33,000 \times g$ for 30 min, and filter the lysate through a 0.22- μm membrane.
7. Apply the filtrate to a 1-mL Strep-Tactin Sepharose (IBA) column pre-equilibrated in resuspension buffer using an ÄKTA Pure (Cytiva) chromatography system.
8. Wash with 20 mL $1 \times$ resuspension buffer, 10 mL $1 \times$ resuspension buffer supplemented with 5 mM ATP, and 10 mL of $1 \times$ resuspension buffer.
9. Elute protein with 15 mL $1 \times$ resuspension buffer supplemented with 2.5 mM desthiobiotin (IBA).
10. Identify the peak fractions by SDS-PAGE and Coomassie staining analysis of 10- μL aliquots of each elution fraction, and pool the peak fractions.
11. Concentrate the protein with an Amicon Ultra-4 30 K centrifugal filter, and load it onto a HiPrep 16/60 Sephacryl S-200 HR column (GE Healthcare Life Sciences) pre-equilibrated in Gel Filtration Buffer.
12. Elute the protein with 1.5 column volumes of Gel Filtration Buffer, and analyze the fractions by SDS-PAGE as above.
13. Pool peak fractions, snap-freeze them with liquid nitrogen, and store at -80°C .

4.1.2 RecQ4 family helicases

Recombinant *S. cerevisiae* Hrq1 was originally generated using *E. coli* (Bochman et al., 2014), but baculovirus-infected insect cell culture has more recently been adopted for its greatly increased yields of protein (Rogers & Bochman, 2017; Rogers et al., 2017). Furthermore, the catalytically inactive Hrq1-K318A mutant and various truncation mutants (e.g., Hrq1 Δ N (Rogers, Lee, et al., 2020)) can be over-expressed and purified likewise, which was recently reviewed (Nickens & Bochman, 2021). Here, we focus on methods to over-express and purify the *Mycobacterium smegmatis* Hrq1 homolog called SftH (Rogers, Lee, et al., 2020).

1. Transform the expression strain of *E. coli* with plasmid pET28a-MsSftH (Rogers, Lee, et al., 2020). We have achieved excellent over-expression with Rosetta 2(DE3) pLysS cells. Select for transformants on LB medium supplemented with the appropriate antibiotics, e.g., 50 μ g/mL kanamycin and 34 μ g/mL chloramphenicol for the above-suggested strains, at 37 °C.
2. Inoculate 10-mL of LB medium supplemented with the same antibiotics with a match head-sized swab of cells from the transformation plate, and grow at 30 °C with aeration for 6–8 h.
3. Inoculate 1 L of ZY-5052 autoinduction medium (Studier, 2005) supplemented with antibiotics with the 10-mL culture, and incubate overnight at 22 °C with aeration. Unless otherwise stated, all subsequent steps should be performed at 4 °C or on ice using instruments and equipment that is likewise chilled.
4. Harvest the cells by pelleting in a centrifuge at $5500 \times g$ for 10 min, decant the growth medium, weigh the pellet(s), store at -80 °C for ≥ 1 h.
5. Resuspend the cells by gently stirring 2 mL $1 \times$ resuspension buffer per gram of cell weight, supplemented with protease inhibitor mix and 10 μ g/mL DNase I.
6. Lyse cells, and clarify the lysate by centrifugation at $33,000 \times g$ for 30 min. We have used a Fisher Scientific FB505 sonicator, a Microfluidics microfluidizer, an Avestin EmulsiFlex-C3 emulsifier, and a high-pressure cell cracker to lyse cells with comparable results, so any lysis method that does not introduce excessive amounts of heat is likely to be acceptable.
7. Apply the supernatant to a 1-mL His60 column, and wash with 10 mL resuspension buffer.

8. Next, wash with 20 mL $1 \times$ resuspension buffer supplemented with 5 mM ATP, followed by 10 mL of $1 \times$ resuspension buffer supplemented with 10 mM imidazole.
9. Elute SftH with $1 \times$ resuspension buffer supplemented with 250 mM imidazole.
10. Identify the peak SftH-containing fractions by SDS-PAGE and Coomassie staining analysis of 10- μ L aliquots of each elution fraction, pool the peak fractions, and dialyze them against a 20-fold excess of Hrql storage buffer.
11. Concentrate the SftH with Amicon Ultra-4 centrifugal filters with a 30,000 cutoff, and store at -80°C in Hrql storage buffer.

4.2 DNA substrate preparation

PIF1 and RecQ helicases are known to interact with a diverse array of nucleic acid substrates *in vivo* and *in vitro* (Bernstein et al., 2010; Bochman et al., 2010), and many of these structures mimicking physiological substrates can be prepared with commercially available oligonucleotides. Here, we describe a simple method to anneal ssDNAs into more complex structures and provide the sequences that can be used to create a variety of model substrates for the biochemical assays listed below.

1. Combine equimolar amounts of the oligonucleotides to be annealed in $1 \times$ Annealing Buffer.
2. Incubate overnight at 37°C .
3. Assess the annealing by running a small portion of the reaction on an 8% 19:1 acrylamide:bisacrylamide native gel in $1 \times$ TG Running Buffer and comparing to the migration of the unannealed labeled oligonucleotide. The gel percentage may need to be optimized to discriminate between the migration of free labeled ssDNA and annealed ssDNA.

If unannealed labeled oligonucleotide is present at $>5\%$ of the total, repeat the annealing process using a slight molar excess of the unlabeled oligonucleotide to drive the incorporation of more of the labeled strand into the desired substrate. The excess amount added should be kept to a minimum to avoid unwanted complications in biochemical assays such as binding to the remaining unlabeled ssDNA instead of the experimental substrate. We have had great luck with the above isothermal annealing method for a host of different substrates (Table 1). However, a more common method is to incubate the annealing reaction in a 95°C heat block for 5 min, turn off the power, and let the block slowly cool to room temperature over several

Table 1 Common DNA substrates used in PIF1 and RecQ experiments.

Name	Sequence (5'-3')	Notes
Poly(dT)5	TTTTT	This series of substrates can be used to gauge the binding site size of helicases using gel shift and, indirectly, ATPase assays (Nickens & Bochman, 2022 ; Rogers et al., 2017). Similar poly-oligonucleotides can be synthesized using any of the other dNTPs and/or with greater length depending on experimental needs
Poly(dT)10	TTTTTTTTTT	
Poly(dT)15	TTTTTTTTTTTTTTTT	
Poly(dT)20	TTTTTTTTTTTTTTTTTTTT	
Poly(dT)25	TTTTTTTTTTTTTTTTTTTTTTTT	
Poly(dT)30	TTTTTTTTTTTTTTTTTTTTTTTTTTTT	
MB733	ACCGTTGTGCAACTGAGTGGACAACGTGTCACTC ACATAGCGTTC	Anneal to generate a DNA fork with 25-nt random-sequence ssDNA arms and a 20-bp duplex
MB734	GAACGCTATGTGAGTGACACCAACAGGTGA GTCAACGTGTTGCCA	
MB1057	CTACTCTAACTCCGACCGCTTGCATGCCTGCAGG	MB1057 and MB1059 form a fork in annealing assays (Rogers & Bochman, 2017). MB1058 can additionally be added in strand exchange assays to convert the fork to blunt dsDNA upon exchange
MB1059	ATCCTCTCTAGAGTCAAGGTCGGAGTTAGAGTAG	
MB1058	CCTGCAGGCATGCAAGCGGTCGGAGTTAGAGTAG	
MB583	GACGCTGCCGAATTCTGGCTTGCTAGGACA TCTTTGCCCACGTTGACCC	These four oligonucleotides can be annealed to create a blunt-ended Holliday junction-like substrate
MB584	TGGGTCAACGTGGGCAAAGATGTCCTAGCA ATGTAATCGTCTATGACGTT	
MB585	CAACGTCATAGACGATTACATTGCTAGGAC ATGCTGTCTAGAGACTATCGA	
MB586	ATCGATAGTCTCTAGACAGCATGTCCTAGC AAGCCAGAATTCGGCAGCGT	

Continued

Table 1 Common DNA substrates used in PIF1 and RecQ experiments.—cont'd

Name	Sequence (5'-3')	Notes
MB808	GGTGAACCTGCAGGTGGGCGGCTGCTCATC GTAGGTTAGTTGGTAGAATTCGGCAGCGTC	Can be annealed to form a ssDNA bubble flanked by blunt dsDNA arms (Rogers et al., 2017)
MB811	GACGCTGCCGAATTCTAGGTTGATTGGATG CTACTCGTCGGCGCCACCTGCAGG TTCACC	
MB1162	GGGTGAACCTGCAGGTGGGCGGCTGCTCAT CGTAGGTTAGTGGTAGAATTCGGCAGCGTC	Can be annealed to create a three-stranded D-loop (Guy & Bolt, 2005)
MB1163	GACGCTGCCGAATTCTACCATGCCTTGCTA GGACATCTTTGCCCACCTGCAGG TTCACCC	
MB1165	TAAGAGCAAGATGTTCTATAAAAGATGTCC TAGCAAGGCA	

hours or, often more conveniently, overnight. In the absence of a heat block, a thermocycler can be utilized in the same way.

4.3 DNA binding assay

Many methods, such as nitrocellulose filter binding (Bochman & Schwacha, 2007) and biolayer interferometry (Ononye, Sausen, et al., 2020), can be used to measure helicase-DNA binding, but here, we describe a simple gel mobility shift assay. Classically, such assays utilized radiolabeled DNA substrates, but we have recently transitioned to using substrates labeled with NIR fluorophores such as the IRDye[®] 700 and 800 modifications available from IDT and others. These dyes can be detected with comparable sensitivity to ³²P but do not require shielding and specialized work areas like radioisotopes and are not subject to the rapid photobleaching in ambient light of the more traditional fluorophores used to label DNA (e.g., fluorescein).

1. Prepare an $n + 10\%$ volume of DNA master mix on ice, where n is the number of reactions to be performed, including a no-protein control. Each reaction requires 4 nM DNA substrate, 5 μ L $2\times$ binding buffer, and Milli-Q H₂O to 10 μ L.
2. On ice, dilute the helicase with $1\times$ binding buffer to the concentrations to be tested. Generally, a range from 1 to 100 nM is a good place to start.
3. Aliquot 10 μ L of DNA master mix into each tube on ice, add 10 μ L of $1\times$ binding buffer (no-protein control) or helicase dilution to each tube, and mix thoroughly by pipetting up and down.
4. Incubate the reactions (e.g., at 30 °C for *S. cerevisiae* proteins or 37 °C for human proteins) for 30 min.
5. Add 4 μ L of $6\times$ nonfluorescent loading buffer to each reaction, and mix thoroughly.
6. Load 20 μ L of each reaction onto an 8% 19:1 acrylamide:bisacrylamide native gel in $1\times$ TG running buffer, and electrophorese at 100 V for 30–45 min.
7. The gels can be imaged with a laser scanning device such as a LI-COR Odyssey DLx and DNA binding quantified with the associated software or freeware such as ImageJ (<https://imagej.nih.gov/ij/>).
8. The percent of DNA bound is calculated by dividing the signal of the shifted DNA by the total DNA signal in the lane and multiplying by 100:

$$DNA\ bound = \frac{Shifted\ DNA}{(Shifted + unshifted\ DNA)} \times 100\%$$

9. DNA binding curves are often hyperbolic for one-site binding or sigmoidal for cooperative binding. These curves can be fitted with software such as GraphPad Prism.

4.4 ATPase assay

This NADH-coupled assay (Fig. 1) (Sausen, Rogers, & Bochman, 2019) has been useful to measure basal and DNA-stimulated ATPase activities of PIF1 (Nickens & Bochman, 2022) and RecQ helicase (Rogers et al., 2017). It can be performed one reaction at a time and monitored spectrophotometrically in standard-sized cuvettes, but here, we present a high-throughput version that can be used to follow up to 96 reactions per run. Although further decreases in reaction size to use less reagents and an increase in the throughput are possible using 384-well plates, pipetting error becomes such a confounding issue that many technical replicates per condition are necessary to average out the noise in the data, negating the throughput and reagent-saving gains. However, the use of liquid handling robotics may be able to overcome this obstacle should it be necessary.

1. Prepare an $n + 10\%$ volume of ATPase reaction master mix on ice, where n is the number of reactions to be performed, including no-protein controls. A single reaction receives 6 μL 100 mM ATP (pH 7.0), 0.6 μL 1 M MgCl_2 , 6 μL 10 mM PEP, 4.6 μL 10 mM NADH, 1.2 μL pyruvate kinase (PK, 10 U/mL; diluted 1:5 with $1 \times$ ATPase Buffer), 1.28 μL lactate dehydrogenase (LDH, 16 U/mL; diluted 1:10 with $1 \times$ ATPase Buffer), and 40.12 μL $1 \times$ ATPase buffer.

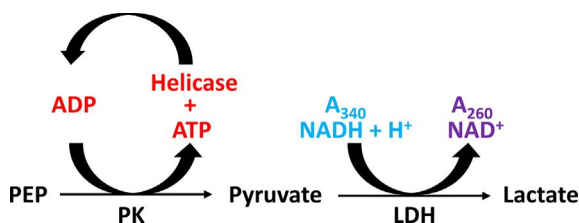


Fig. 1 NADH-coupled ATPase assay. In this assay, a helicase (+/– DNA, not shown) binds and hydrolyzes ATP into ADP. To maintain the ATP at a constant concentration, phosphoenol pyruvate (PEP) and pyruvate kinase (PK) are used as an ATP regeneration system, recycling ADP into ATP and generating pyruvate. This reaction is coupled to lactate production by the action of lactate dehydrogenase (LDH) on pyruvate in the presence of NADH. The conversion of NADH to NAD⁺ is monitored by recording the decrease in absorbance at 340 nm (A_{340}). One ATP is hydrolyzed for every NADH oxidized.

2. In a separate tube on ice, dilute the helicase of interest to 10 nM in $1 \times$ ATPase buffer. Make this as a protein master mix with enough volume for 60 μL per well.
3. Chill a 96-well plate on ice.
4. Pipet 60 μL of the protein master mix into each well of the 96-well plate to be monitored. Similarly, add 60 μL of $1 \times$ ATPase Buffer to the no-protein control well(s).
5. If assessing the effects of DNA on the ATP hydrolysis rate of the helicase, add 1.2 μL of 1 μM DNA substrate (ssDNA, dsDNA, fork, etc.) to each well or an equal volume of buffer to no-DNA wells. Some autohydrolysis of ATP also occurs, so an ATP-only control lacking added helicase is also advisable.
6. Add 60 μL of ATPase reaction master mix to each well, and mix by pipetting up and down. A multichannel pipette can be used to simultaneously mix many wells at once.
7. Wipe off the bottom of the wells to remove ice and moisture, place the plate in the plate reader (a BioTek Synergy H1 or similar instrument), and monitor absorbance at 340 nm at the desired temperature every 10 min for 1 h. Only the linear portion of the data is used to calculate the rate of hydrolysis (i.e., the slope of the line) in $\mu\text{M}/\text{min}$. For very fast ATPases, more frequent measurements may be needed to capture the linear portion of the reaction, or a lower reaction temperature can be used to slow the rate of hydrolysis.
8. Absorbance readings are converted to ATP turnover (μM) based on the NADH concentration. It is assumed that 1 μM NADH oxidized is proportional to 1 μM ATP hydrolyzed.

NADH and PEP are consumed in a 1:1 ratio during the reaction to produce a change in A_{340} . By doubling their concentrations, the linear range of the reaction can be doubled, which is useful for fast reactions. If a reaction ends (i.e., runs out of NADH) in <10 min, it may be approaching the maximal rate of the PK and LDH enzymes, and therefore, they may become rate limiting. In this case, the reaction needs to be adjusted to increase the time domain. If the reaction cannot be slowed, the PK and LDH enzyme concentrations can be increased to compensate. If NADH and PEP are doubled, then reactions ending in 20 min may be approaching the rate limit of PK and LDH, and the same adjustments need to be made.

When adding ssDNA to the reactions, secondary structures may cause confounding results. To minimize these effects, the ssDNA oligonucleotides should be diluted to their working concentration(s), boiled, and snap-cooled prior to addition to the ATPase assays.

4.5 Helicase assay

As with DNA binding, there are many ways to monitor DNA unwinding (e.g., Ononye, Sausen, et al., 2020; Zhou, Zhang, Bochman, Zakian, & Ha, 2014), but we again present a simple gel-based method (Rogers, Sausen, & Bochman, 2019) using NIR-labeled substrates for its simplicity. All manipulations should be performed on ice unless otherwise stated.

1. Dilute recombinant helicase from freezer stocks to fivefold the desired final concentration using $1 \times$ binding buffer. For Pif1, concentrations of ≤ 100 nM are generally appropriate, but for less active enzymes like Hrq1, up to 500 nM may be required to reach maximal activity.
2. Mix 2 μ L of helicase with 1 μ L 10 nM DNA substrate, 0.5 μ L 100 mM ATP, 1 μ L H₂O, and 5 μ L $2 \times$ binding buffer. For proteins that exhibit strong annealing activity, such as human RECQL4 and *S. cerevisiae* Pif1, add 10 nM cold trap DNA (i.e., a molar excess of unlabeled ssDNA equivalent to the labeled strand in your substrate). Two reactions should omit protein: one for use as a negative control to measure spontaneous DNA dissociation during the reaction; and the other for a boiled control (Fig. 2; see below).
3. Incubate the reactions (e.g., at 30 °C for *S. cerevisiae* proteins or 37 °C for human proteins) for 30 min, and then place them on ice.
4. To each, add 5 μ L of $3 \times$ nonfluorescent loading buffer supplemented with 400 μ g/mL SDS-Proteinase K, and incubate the reactions at 37 °C for 10 min to release protein from the DNA.
5. Heat the boiled control to 95 °C for 5 min, and snap-cool on ice to use as a reference for the migration of fully unwound substrate.
6. Load 10 μ L of the reactions on an 8% 19:1 acrylamide:bisacrylamide gel using $1 \times$ TBE running buffer, and run at 10 V/cm until the dye front reaches the bottom of the gel.
7. The gels can be imaged as described above for DNA binding gel shifts. The percent of DNA unwound can be calculated by dividing the ssDNA signal by the total signal in each lane, and the amount of ssDNA in the no-protein control should be subtracted from each of the reactions:

$$DNA\ unwound = \left(\left(\frac{ssDNA}{ssDNA + dsDNA} \right) - (spontaneous\ ssDNA) \right) \times 100\%$$

8. DNA unwinding curves are usually hyperbolic and can be fitted with software such as GraphPad Prism.

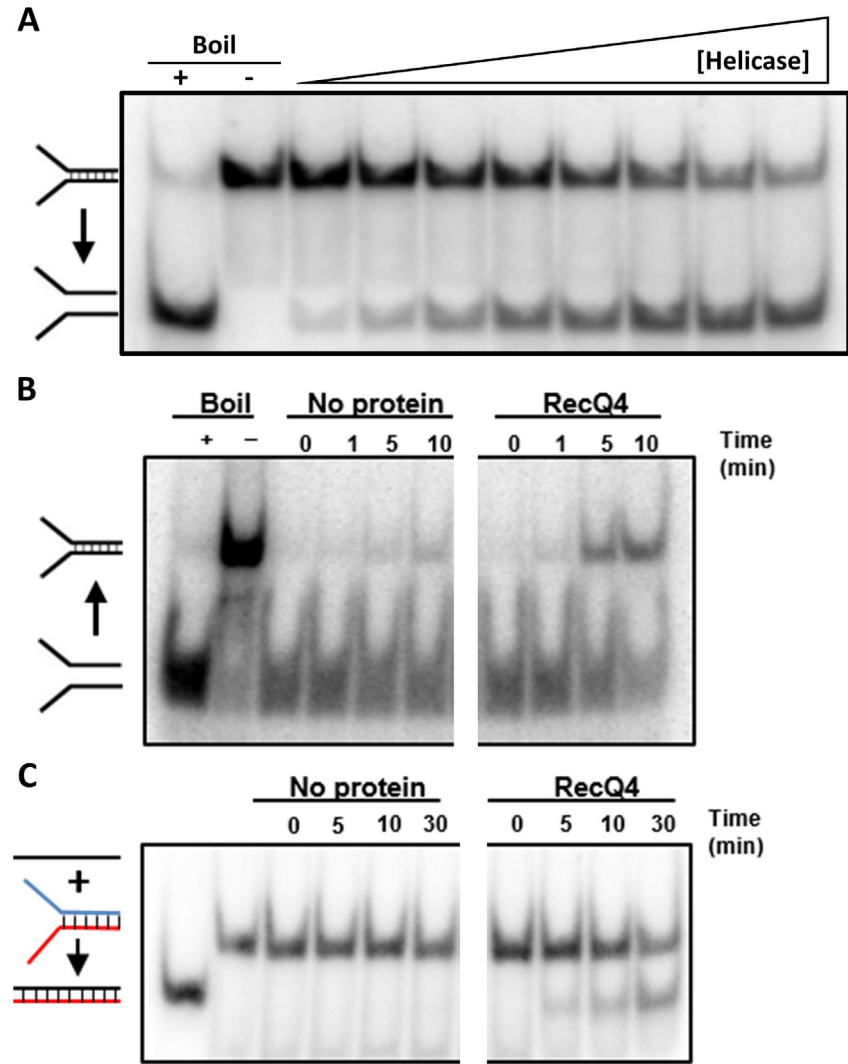


Fig. 2 Example helicase, DNA annealing, and stand exchange assays. (A) Conversion of the fork substrate into ssDNA by unwinding the duplex portion of the fork substrate is catalyzed by the addition of a DNA helicase in a concentration-dependent manner (1–100nM). The boiled control (+) shows the position at which the ssDNA should migrate through the gel, and the unboiled control (–) shows where the fork substrate migrates ($t = 30$ min endpoints for each control). The fork substrate shown was created by annealing oligonucleotides MB1057 and MB1058 (Table 1). (B) Annealing of two partially complementary ssDNAs (MB1057 and MB1058) into a DNA fork by the RECQL4 (100nM; RecQ4) helicase. Here, the annealed product (i.e., the fork) was boiled (+) to indicate the position where the unannealed ssDNAs migrate through the gel, and the unboiled (–) control indicates the migration position of the annealed fork. (C) Strand exchange catalyzed by 50nM RECQL4. In this assay, the unlabeled partially complementary oligonucleotide (blue, MB1058) that forms half of the fork substrate (Continued)

4.6 DNA annealing assay

DNA annealing is the opposite of helicase activity, that is, the annealing of two ssDNAs into a dsDNA substrate. Many PIF1 and RecQ family helicases display this activity (Lu & Davis, 2021; Ramanagoudr–Bhojappa, Byrd, Dahl, & Raney, 2014), though there are some species-specific differences among homologs, such as Hrq1 and RECQL4 (Rogers & Bochman, 2017). Regardless, the assay is quite similar to the helicase assay described above, but it omits ATP to avoid the confounding effects of the helicase cycling between annealing and unwinding modes.

1. On ice, dilute helicase protein from freezer stocks to fivefold the desired final concentration (10–100 nM) using $1 \times$ binding buffer.
2. Prepare the complementary ssDNAs by heat-denaturing a helicase assay substrate such as a fork for 10 min at 95 °C and snap-cooling on ice.
3. Mix 2 μ L helicase, 1 μ L of 10 μ M complementary ssDNAs, 2 μ L Milli-Q H₂O, and 5 μ L $2 \times$ binding buffer. Omit protein from one reaction to use as a negative control for spontaneous annealing. Another reaction should likewise contain no protein but an equimolar amount of annealed substrate (i.e., that was not heat-denatured in step 2) as a control for where the fully annealed substrate should migrate.
4. Incubate the reactions (e.g., at 30 °C for *S. cerevisiae* proteins or 37 °C for human proteins) for 30 min.
5. Stop the reactions by adding 5 μ L of $3 \times$ nonfluorescent loading buffer supplemented with 400 μ g/mL SDS-Proteinase K to each, and incubate the reactions at 37 °C for 10 min to release protein from the DNA.
6. The gels can be imaged as described above for DNA binding gel shifts. The percent of annealing can be calculated by dividing the annealed signal by the total signal in each lane, and the amount of spontaneously annealed product in the no-protein control should be subtracted from each of the reactions:

$$\text{Annealed DNA} = \left(\left(\frac{\text{dsDNA}}{\text{dsDNA} + \text{ssDNA}} \right) - (\text{spontaneous dsDNA}) \right) \times 100\%$$

Fig. 2—Cont'd in (A) and (B) can be exchanged for a completely complementary unlabeled oligonucleotide (black, MB1059) to produce blunt dsDNA, which migrates more quickly in the gel than the fork. The images in (B) and (C) are adapted from Rogers, C. M., & Bochman, M. L. (2017). *Saccharomyces cerevisiae* Hrq1 helicase activity is affected by the sequence but not the length of single-stranded DNA. *Biochemical and Biophysical Research Communications*, 486(4), 1116–1121. doi:10.1016/j.bbrc.2017.04.003.

4.7 Strand exchange assay

Some helicases that display annealing activity are additionally capable of strand exchange (Keller et al., 2014), i.e., the trading of one strand of a dsDNA duplex for a complementary ssDNA, similar to what occurs during strand exchange for homologous recombination (Daley, Kwon, Niu, & Sung, 2013). In general, substrates for this assay must be designed such that the annealed DNA structure has a different migration pattern than the strand exchange product on a native gel. In this example, it is the conversion of a fork substrate to a blunt dsDNA substrate. Alternatively, two different NIR labels can be used to discriminate between the original substrate and the strand exchange product.

1. On ice, dilute helicase protein from freezer stocks to fivefold the desired final concentration using $1 \times$ binding buffer.
2. Mix 2 μ L helicase, 5 μ L $2 \times$ Binding Buffer, 1 μ L Milli-Q H_2O , 1 μ L of 10 nM substrate (e.g., a DNA fork), and 1 μ L of 10 nM complementary ssDNA. Omit protein from one reaction to use as a negative control for spontaneous strand exchange. Another reaction should likewise contain no protein but an equimolar amount of dsDNA substrate as a control for where the fully strand exchanged substrate should migrate.
3. Incubate the reactions (e.g., at 30 °C for *S. cerevisiae* proteins or 37 °C for human proteins) for 30 min.
4. Stop the reactions by adding 5 μ L of $3 \times$ nonfluorescent loading buffer supplemented with 400 μ g/mL SDS-Proteinase K to each and incubate the reactions at 37 °C for 10 min to release protein from the DNA.
5. The gels can be imaged as described above for DNA binding gel shifts. The percent of strand exchange can be calculated by dividing the dsDNA signal by the total signal in each lane:

$$\text{Strand exchange} = \frac{\text{Blunt dsDNA}}{\text{Blunt dsDNA} + \text{Fork DNA}} \times 100\%$$

References

- Andis, N. M., Sausen, C. W., Alladin, A., & Bochman, M. L. (2018). The WYL domain of the PIF1 helicase from the thermophilic bacterium *Thermotoga elfii* is an accessory single-stranded DNA binding module. *Biochemistry*, 57(7), 1108–1118. <https://doi.org/10.1021/acs.biochem.7b01233>.
- Bachrati, C. Z., & Hickson, I. D. (2008). RecQ helicases: Guardian angels of the DNA replication fork. *Chromosoma*, 117, 219–233. Retrieved from http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=18188578.

- Berger, J. M. (2008). SnapShot: Nucleic acid helicases and translocases. *Cell*, 134. <https://doi.org/10.1016/j.cell.2008.08.027>. 888–888.e881.
- Bernstein, K. A., Gangloff, S., & Rothstein, R. (2010). The RecQ DNA helicases in DNA repair. *Annual Review of Genetics*, 44, 393–417. <https://doi.org/10.1146/annurev-genet-102209-163602>.
- Bessler, J. B., & Zakian, V. A. (2004). The amino terminus of the *Saccharomyces cerevisiae* DNA helicase Rrm3p modulates protein function altering replication and checkpoint activity. *Genetics*, 168, 1205–1218.
- Bochman, M. L. (2014). Roles of DNA helicases in the maintenance of genome integrity. *Molecular & Cellular Oncology*, 1(3), e963429. <https://doi.org/10.4161/23723548.2014.963429>.
- Bochman, M. L., Judge, C. P., & Zakian, V. A. (2011). The Pif1 family in prokaryotes: What are our helicases doing in your bacteria? *Molecular Biology of the Cell*, 22(12), 1955–1959. <https://doi.org/10.1091/mbc.E11-01-0045>.
- Bochman, M. L., Paeschke, K., Chan, A., & Zakian, V. A. (2014). Hrq1, a homolog of the human RecQ4 helicase, acts catalytically and structurally to promote genome integrity. *Cell Reports*, 6(2), 346–356. <https://doi.org/10.1016/j.celrep.2013.12.037>.
- Bochman, M. L., Sabouri, N., & Zakian, V. A. (2010). Unwinding the functions of the Pif1 family helicases. *DNA Repair (Amst)*, 9, 237–249. <https://doi.org/10.1016/j.dnarep.2010.01.008>.
- Bochman, M. L., & Schwacha, A. (2007). Differences in the single-stranded DNA binding activities of MCM2–7 and MCM467: MCM2 and MCM5 define a slow ATP-dependent step. *The Journal of Biological Chemistry*, 282, 33795–337804. <https://doi.org/10.1074/jbc.M703824200>.
- Bohr, V. A. (2008). Rising from the RecQ-age: The role of human RecQ helicases in genome maintenance. *Trends in Biochemical Sciences*, 33, 609–620. <https://doi.org/10.1016/j.tibs.2008.09.003>.
- Chu, W. K., & Hickson, I. D. (2009). RecQ helicases: Multifunctional genome caretakers. *Nature Reviews. Cancer*, 9, 644–654. <https://doi.org/10.1038/nrc2682>.
- Daley, J. M., Kwon, Y., Niu, H., & Sung, P. (2013). Investigations of homologous recombination pathways and their regulation. *The Yale Journal of Biology and Medicine*, 86(4), 453–461. Retrieved from <https://www.ncbi.nlm.nih.gov/pubmed/24348209>.
- Foury, F., & Dyck, E. V. (1985). A PIF-dependent recombinogenic signal in the mitochondrial DNA of yeast. *The EMBO Journal*, 4, 3525–3530. Retrieved from http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=16453651.
- Gupta, S. V., & Schmidt, K. H. (2020). Maintenance of yeast genome integrity by RecQ family DNA helicases. *Genes (Basel)*, 11(2), 205. <https://doi.org/10.3390/genes11020205>.
- Guy, C. P., & Bolt, E. L. (2005). Archaeal Hel308 helicase targets replication forks in vivo and in vitro and unwinds lagging strands. *Nucleic Acids Research*, 33(11), 3678–3690. <https://doi.org/10.1093/nar/gki685>.
- Hall, M. C., & Matson, S. W. (1999). Helicase motifs: The engine that powers DNA unwinding. *Molecular Microbiology*, 34, 867–877. Retrieved from <http://www.ncbi.nlm.nih.gov/htbin-post/Entrez/query?db=m&form=6&dopt=r&uid=0010594814>.
- Keller, H., Kiosze, K., Sachsenweger, J., Haumann, S., Ohlenschläger, O., Nuutinen, T., et al. (2014). The intrinsically disordered amino-terminal region of human RecQL4: Multiple DNA-binding domains confer annealing, strand exchange and G4 DNA binding. *Nucleic Acids Research*, 42(20), 12614–12627. <https://doi.org/10.1093/nar/gku993>.
- Kornberg, A. (2003). Ten commandments of enzymology, amended. *Trends in Biochemical Sciences*, 28(10), 515–517. <https://doi.org/10.1016/j.tibs.2003.08.007>.

- Lu, H., & Davis, A. J. (2021). Human RecQ helicases in DNA double-Strand break repair. *Frontiers in Cell and Development Biology*, 9, 640755. <https://doi.org/10.3389/fcell.2021.640755>.
- Muellner, J., & Schmidt, K. H. (2020). Yeast genome maintenance by the multifunctional Pif1 DNA helicase family. *Genes (Basel)*, 11(2), 224. <https://doi.org/10.3390/genes11020224>.
- Nickens, D. G., & Bochman, M. L. (2021). Characterization of the telomerase modulating activities of yeast DNA helicases. *Methods in Enzymology*, 661, 327–342. <https://doi.org/10.1016/bs.mie.2021.08.005>.
- Nickens, D. G., & Bochman, M. L. (2022). Pif1 activity is modulated by DNA sequence and structure. *Biochemistry*, 61(1), 10–20. <https://doi.org/10.1021/acs.biochem.1c00614>.
- Nickens, D. G., Rogers, C. M., & Bochman, M. L. (2018). The *Saccharomyces cerevisiae* Hrq1 and Pif1 DNA helicases synergistically modulate telomerase activity in vitro. *The Journal of Biological Chemistry*, 293(37), 14481–14496. <https://doi.org/10.1074/jbc.RA118.004092>.
- Nickens, D. G., Sausen, C. W., & Bochman, M. L. (2019). The biochemical activities of the *Saccharomyces cerevisiae* Pif1 helicase are regulated by its N-terminal domain. *Genes (Basel)*, 10(6), 411. <https://doi.org/10.3390/genes10060411>.
- Ononye, O. E., Sausen, C. W., Balakrishnan, L., & Bochman, M. L. (2020). Lysine acetylation regulates the activity of nuclear Pif1. *The Journal of Biological Chemistry*, 295(46), 15482–15497. <https://doi.org/10.1074/jbc.RA120.015164>. Epub 2020 Sep 2.
- Ononye, O. E., Sausen, C. W., Bochman, M. L., & Balakrishnan, L. (2021). Dynamic regulation of Pif1 acetylation is crucial to the maintenance of genome stability. *Current Genetics*, 67(1), 85–92. <https://doi.org/10.1007/s00294-020-01116-5>. Epub 2020 Oct 20.
- Ramanagoudr-Bhojappa, R., Byrd, A. K., Dahl, C., & Raney, K. D. (2014). Yeast Pif1 accelerates annealing of complementary DNA strands. *Biochemistry*, 53(48), 7659–7669. <https://doi.org/10.1021/bi500746v>.
- Rogers, C. M., & Bochman, M. L. (2017). *Saccharomyces cerevisiae* Hrq1 helicase activity is affected by the sequence but not the length of single-stranded DNA. *Biochemical and Biophysical Research Communications*, 486(4), 1116–1121. <https://doi.org/10.1016/j.bbrc.2017.04.003>.
- Rogers, C. M., Lee, C. Y., Parkins, S., Buehler, N. J., Wenzel, S., Martinez-Marquez, F., et al. (2020). The yeast Hrq1 helicase stimulates Pso2 translesion nuclease activity and thereby promotes DNA interstrand crosslink repair. *The Journal of Biological Chemistry*, 295(27), 8945–8957. <https://doi.org/10.1074/jbc.RA120.013626>.
- Rogers, C. M., Sanders, E., Nguyen, P. A., Smith-Kinnaman, W., Mosley, A. L., & Bochman, M. L. (2020). The genetic and physical Interactomes of the *Saccharomyces cerevisiae* Hrq1 helicase. *G3 (Bethesda)*, 10(12), 4347–4357. <https://doi.org/10.1534/g3.120.401864>.
- Rogers, C. M., Sausen, C. W., & Bochman, M. L. (2019). Gel-based assays for measuring DNA unwinding, annealing, and strand exchange. *Methods in Molecular Biology*, 1999, 255–264. https://doi.org/10.1007/978-1-4939-9500-4_16.
- Rogers, C. M., Simmons, R. H., III, Fluhler Thornburg, G. E., Buehler, N. J., & Bochman, M. L. (2020). Fanconi anemia-independent DNA inter-strand crosslink repair in eukaryotes. *Progress in Biophysics and Molecular Biology*, 158, 33–46. <https://doi.org/10.1016/j.pbiomolbio.2020.08.005>. Epub 2020 Aug 30.
- Rogers, C. M., Wang, J. C., Noguchi, H., Imasaki, T., Takagi, Y., & Bochman, M. L. (2017). Yeast Hrq1 shares structural and functional homology with the disease-linked human RecQ4 helicase. *Nucleic Acids Research*, 45(9), 5217–5230. <https://doi.org/10.1093/nar/gkx151>.

- Rossi, S. E., Ajazi, A., Carotenuto, W., Foiani, M., & Giannattasio, M. (2015). Rad53-mediated regulation of Rrm3 and Pif1 DNA helicases contributes to prevention of aberrant fork transitions under replication stress. *Cell Reports*, 13(1), 80–92. <https://doi.org/10.1016/j.celrep.2015.08.073>.
- Sanders, E., Nguyen, P. A., Rogers, C. M., & Bochman, M. L. (2020). Comprehensive synthetic genetic array analysis of alleles that interact with mutation of the *Saccharomyces cerevisiae* RecQ helicases Hrq1 and Sgs1. *G3 (Bethesda)*, 10(12), 4359–4368. <https://doi.org/10.1534/g3.120.401709>.
- Sausen, C. W., Rogers, C. M., & Bochman, M. L. (2019). Thin-layer chromatography and real-time coupled assays to measure ATP hydrolysis. *Methods in Molecular Biology*, 1999, 245–253. https://doi.org/10.1007/978-1-4939-9500-4_15.
- Simmons, R. H., 3rd, Rogers, C. M., & Bochman, M. L. (2021). A deep dive into the RecQ interactome: Something old and something new. *Current Genetics*, 67(5), 761–767. <https://doi.org/10.1007/s00294-021-01190-3>.
- Studier, F. W. (2005). Protein production by auto-induction in high density shaking cultures. *Protein Expression and Purification*, 41(1), 207–234. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/15915565>.
- Stundon, J. L., & Zakian, V. A. (2015). Identification of *Saccharomyces cerevisiae* genes whose deletion causes synthetic effects in cells with reduced levels of the nuclear Pif1 DNA helicase. *G3 (Bethesda)*, 5(12), 2913–2918. <https://doi.org/10.1534/g3.115.021139>.
- Vasianovich, Y., Harrington, L. A., & Makovets, S. (2014). Break-induced replication requires DNA damage-induced phosphorylation of Pif1 and leads to telomere lengthening. *PLoS Genetics*, 10(10), e1004679. <https://doi.org/10.1371/journal.pgen.1004679>.
- Zhou, R., Zhang, J., Bochman, M. L., Zakian, V. A., & Ha, T. (2014). Periodic DNA patrolling underlies diverse functions of Pif1 on R-loops and G-rich DNA. *eLife*, 3, e02190. <https://doi.org/10.7554/eLife.02190>.
- Zybailov, B., Gokulan, K., Wiese, J., Ramanagoudr-Bhojappa, R., Byrd, A. K., Glazko, G., et al. (2015). Analysis of protein-protein interaction Interface between yeast mitochondrial proteins Rim1 and Pif1 using chemical cross-linking mass spectrometry. *The Journal of Proteomics & Bioinformatics*, 8(11), 243–252. <https://doi.org/10.4172/jpb.1000376>.